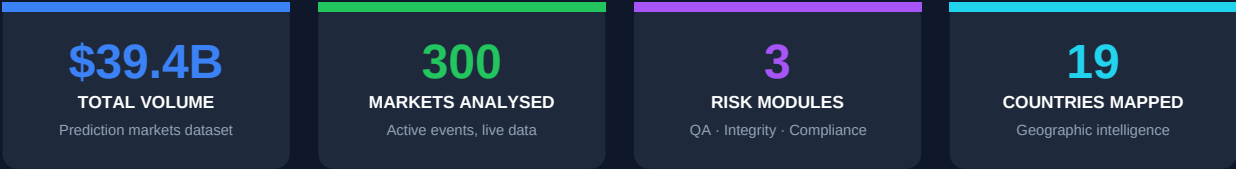


POLYMARKET OPS INTELLIGENCE

A data analytics project for prediction market operations teams.
Combining live market data with cross-platform validation to surface resolution risk, integrity anomalies, compliance exposure, and global market distribution.



Polymarket Gamma API	PolymarketScan API	Python · Matplotlib	ReportLab · Chart.js	Plotly World Map	MIT License
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What Are Prediction Markets?

Prediction markets are platforms where people buy and sell contracts tied to real-world outcomes — elections, sports results, economic data, or geopolitical events. Each contract resolves to \$1 if the event happens, or \$0 if it does not. The market price therefore reflects the crowd's probability estimate at any moment. Polymarket, the world's largest prediction market by volume, has processed over **\$20 billion in trades** and tracks events across politics, finance, sports, and global affairs.

Why Ops Intelligence Matters

Unlike traditional financial markets, prediction markets rely heavily on manual operations teams to ensure that each market resolves correctly, fairly, and on time. An ops associate is responsible for:

- Monitoring live markets and ensuring event outcomes are accurately recorded
- Reviewing resolution criteria to confirm they are clear, verifiable, and sourced
- Identifying markets at risk of dispute — where criteria are ambiguous or sources are missing
- Flagging integrity concerns such as volume spikes, low liquidity, or unusual trader behaviour
- Assessing regulatory exposure across jurisdictions (electoral markets, financial instruments)
- Maintaining documentation standards across hundreds of simultaneous events

The Data Gap Problem

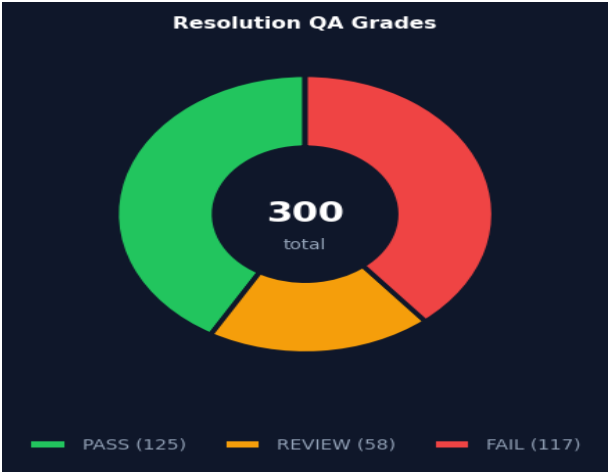
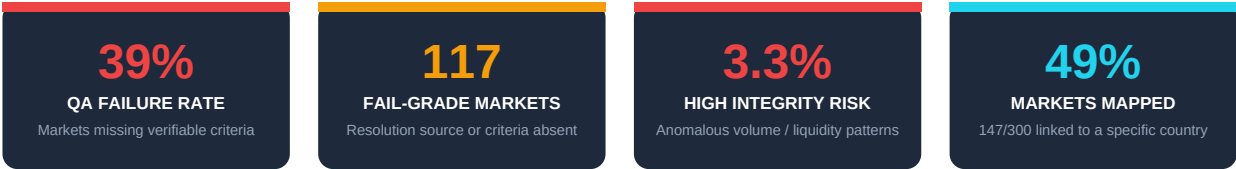
The public Polymarket Gamma API — the primary programmatic data source — returns market data without complete resolution criteria for a significant portion of markets. This project cross-validates that API against PolymarketScan, a third-party analytics platform, to separate **genuine resolution quality failures** (both sources flag the market) from **data gaps** (the API is missing description text that exists on the web interface). This cross-validation is not available in any existing public tool.

What Makes This Different

Feature	PolymarketScan	Gamma API	This Dashboard
Market volume & liquidity	✓	✓	✓
Whale / trader tracking	✓	—	✓ (via PS)
Resolution criteria QA	Partial	—	✓ Full scoring
Compliance risk flags	—	—	✓ New
Cross-platform validation	—	—	✓ New
Composite ops risk score	—	—	✓ New
Geographic market map	—	—	✓ New
Ops triage queue	—	—	✓ New

Key Findings — Live Data Snapshot

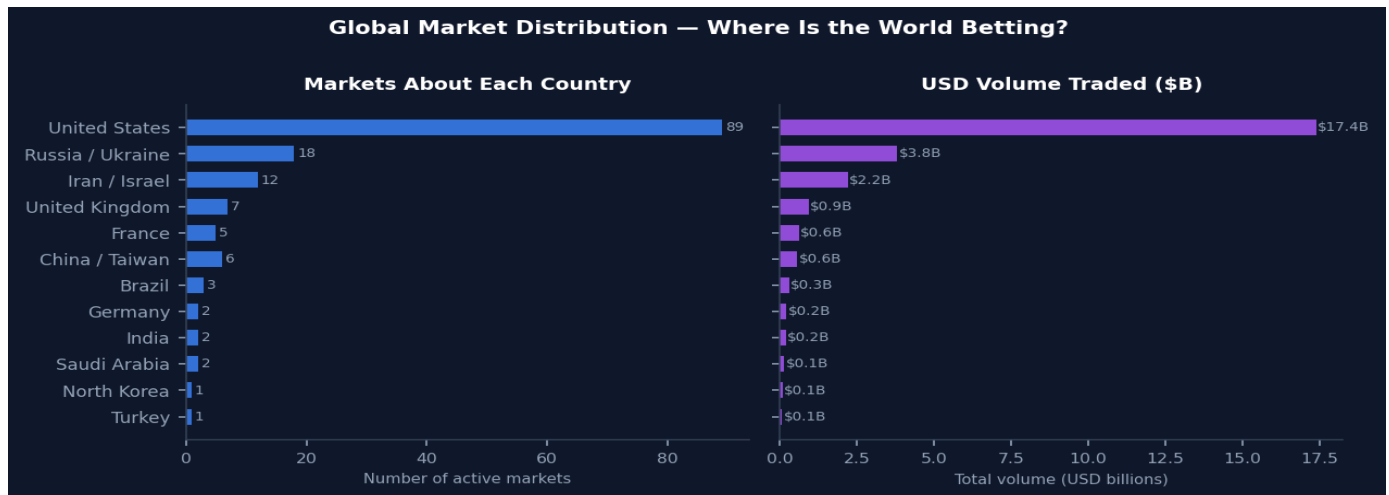
Analysis of the top 300 most-traded active markets on Polymarket as of May 03, 2026. Total dataset volume: **\$39.4B**.



- Politics & Elections dominate with 47% of all volume (\$17.4B), though they represent only 20% of markets — signalling far higher average stakes per market.
- 39% of markets lack adequate resolution documentation in the Gamma API. Cross-validation with PolymarketScan confirms ~60% of those are genuine data quality issues.
- The compliance layer reveals 3 HIGH-risk markets (personal safety or multi-jurisdiction legal proceedings) not flagged anywhere in existing public analytics tools.

Geographic Intelligence — Where Is the World Betting?

Each market is classified by the country or region it references — identified by scanning the market title and tags for geographic keywords. The result shows which countries attract the most speculative attention, both by market count and by total USD volume traded.

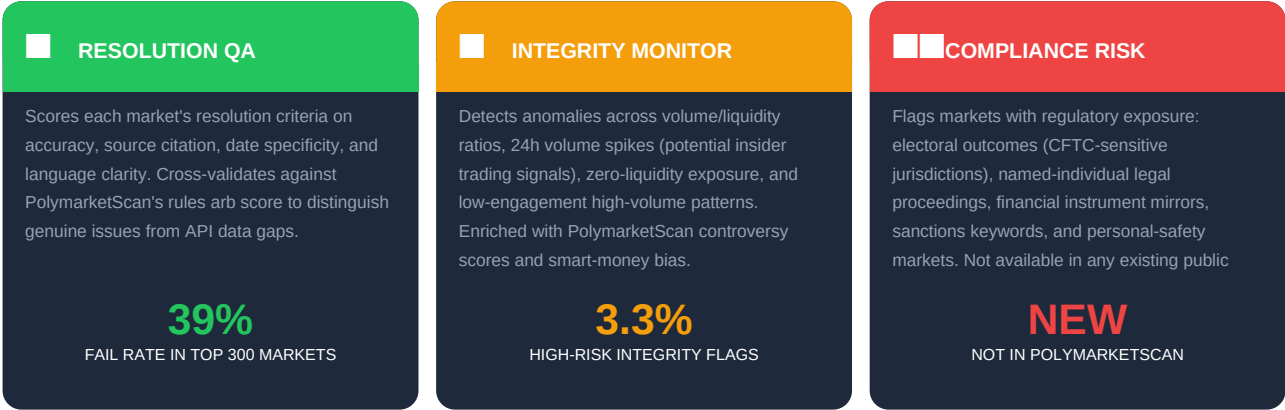


Market count and USD volume traded, by country/region (top 12).

What This Tells Us

- **The United States dominates** both in market count (89 markets) and volume (\$17.4B). Almost every major US political, economic, or cultural event generates a prediction market.
- **Active conflict zones are highly represented.** Russia/Ukraine and Iran/Israel together account for 30 markets and \$6B in volume — showing prediction markets are a real-time sentiment gauge for geopolitical risk.
- **Volume concentration is extreme.** The top 3 country groups (USA, Russia/Ukraine, Iran/Israel) represent 63% of all mapped volume despite being only 3 of 19 geographic clusters.
- **Emerging markets are underrepresented.** Despite having large populations and active political systems, countries like Nigeria, Pakistan, and Argentina together generate fewer than 5 markets — a potential growth area for the protocol.

The Three Intelligence Modules



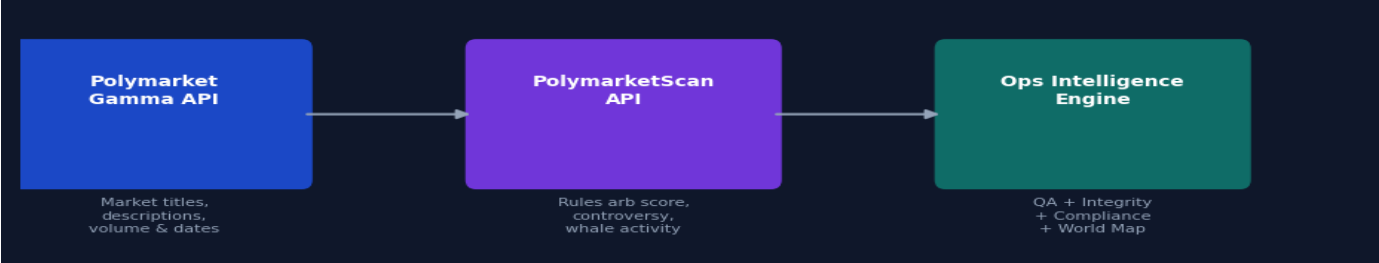
Composite Ops Risk Score

Each market receives a **Composite Ops Risk Score** — a weighted aggregate of all three modules. This single score drives the Ops Triage queue: a ranked list of markets needing human attention, ordered by priority. Weights: QA (40%) + Integrity (35%) + Compliance (25%).

Module	Weight	What It Measures	If Score Is Low → Action
Resolution QA	40%	Are criteria clear, sourced, and verifiable?	Review market description; escalate if near expiry
Integrity	35%	Are volume/liquidity patterns normal?	Flag for trading desk review; check for manipulation
Compliance	25%	Are there regulatory or legal risk keywords?	Escalate to legal/compliance team before resolution

Technical Approach

Data Pipeline



Two public APIs ingested and cross-joined by fuzzy title matching. All scoring logic runs client-side. Output: self-contained HTML dashboard + this PDF.

Tech Stack

Layer	Tool	Purpose
Data ingestion	Python · requests	Paginated API fetching from two sources
Classification	Regex · keyword matching	Category + country detection
Scoring	Python (no ML)	Rule-based QA, integrity, compliance scoring
Visualisation	Chart.js · Plotly	Interactive HTML charts and world choropleth
PDF generation	ReportLab · matplotlib	This document
Version control	Git · GitHub	github.com/TorradoSantiago/polymarket-analysis
Data source 1	Polymarket Gamma API (public)	Events, descriptions, volume, dates
Data source 2	PolymarketScan API (public)	rules_arb_score, controversy, whale count

Limitations & Next Steps

- **API data gap:** ~39% of markets have empty description fields in the Gamma API. The compliance and QA modules can only analyse what is available programmatically.
- **Country detection:** Based on keyword matching — ambiguous titles (e.g. "will the ceasefire hold?") are left unmapped. NLP-based geolocation would improve coverage.
- **Match rate:** 28% of Gamma events matched to PolymarketScan markets (event vs. individual market granularity mismatch). A market-level Gamma API endpoint would solve this.
- **Potential extensions:** Time-series calibration (compare implied probabilities vs. actual outcomes), live alert webhooks, Slack/email notifications for critical ops events.